

Mohamed Tlili

GIS Consultant - Mineral Exploration

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Geomatics engineer specializing in GIS, cartography, and LiDAR processing, with expertise in geodatabase design, python automation, and spatial data production for mineral exploration.



Professional Experience

EAGLEPROJECTS TUNISIA (a subsidiary of EAGLEPROJECTS S.p.A., Italy) - Geomatic Specialist

Jul 2026 - Present

- Processed and registered point clouds from laser scanning and/or photogrammetric surveys.
- Cleaned, filtered, and optimized datasets.
- Aligned and georeferenced point clouds.
- Created 3D models, meshes, and surfaces from acquired data.
- Ensured structured management and storage of datasets.
- Collaborated with design teams (BIM, CAD, GIS).

WOLO Engineering | Ariana, Tunisia | Dec 2023 - Present

Team Lead

Dec 2025 - Dec 2026

- Led a team designing and deploying a cartographic geodatabase in ArcGIS Pro with custom data dictionaries; built automated data-validation pipelines (Python + geodatabase rules) and integrated restitution deliverables for specification compliance.
- Coordinated project delivery and deadline adherence, ensuring team cohesion and continuous alignment with client requirements.

Cartographic Study Specialist

Dec 2023 - Dec 2025

- Processed and classified LiDAR point clouds (ground, vegetation, buildings) using MicroStation and TerraScan; generated digital terrain models DTM/DSM and performed noise filtering.
- Conducted GCP placement and dense point-cloud processing in Agisoft Metashape; generated DTM/DSM and produced high-resolution orthomosaics.
- Updated fiber-optic and infrastructure data in Smallworld GE; coordinated with cross-functional teams and clients to align spatial data with AutoCAD plans.
- Conducted precise digitization and feature extraction from drone imagery, ensuring the accuracy and reliability of cartographic deliverables.

GIS Consultant Freelance - Mineral Exploration

Nov 2025 - Present

- Produced thematic maps, cartographic layouts, and technical documents for land analysis, mineral tenure, and geological exploration projects.
- Identified staking opportunities and detected claim overlaps by cross-referencing geological, cadastral, and tenure datasets (PLSS, BLM, Crown concessions, LTSA, mining claims) for targeted mineral exploration.
- Converted GIS data to CAD and created professional AutoCAD plans with complete cartographic styling, managing coordinate transformations and datum conversions in ArcGIS Pro.
- Automated repetitive geoprocessing and file management tasks using Python; built custom ArcGIS toolboxes to streamline spatial data workflows.

Skills

GIS & Cartography: ArcGIS Pro, QGIS (advanced), Smallworld GIS, Agisoft Metashape | **CAD:** AutoCAD, MicroStation, FME | **Programming:** Python (geoprocessing, automation, toolboxes), LAS tools

Projects

- **LAS File Manager:** Developed a Python application to split and merge LAS files by X/Y axes or grid, with interactive HTML map preview and real georeferencing; automated workflow using LAStools for significant time savings in LiDAR data preparation.
- **Geodatabase Automation:** Automated batch import of 200+ CSV/Excel files into a geodatabase with field mapping and data normalization; built dynamic spatial attribute updates between shapefiles.
- **Geological Classification Tool:** Automated lithology and geological unit classification into 10 standard categories, ensuring data consistency and streamlined analysis.
- **UTM Grid Generator:** Generated UTM coordinate grids in AutoCAD for multiple coordinate systems and projections using a custom LISP script.

Education

National Engineering Diploma - Geomatics and Topography

Ecole Supérieure de l'Aéronautique et des Technologies | Ariana, Tunisia | 2020-2023 | EQF Level 7

Language Skills

Arabic (Native) | **French** (Professional) | **English** (Professional)